A comprehensive semantic analysis for *doubt* in English and its extension to the Mandarin dubitative predicate 怀疑(huái yí)

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1 Introduction

The semantic analysis for dubitative predicates has always been a challenging research topic. One aspect of the challenge is about defining a precise semantic interpretation, while the other is that they present peculiar selectional restrictions regarding embedding complements: Cross-linguistically, the pattern is that they are compatible with declarative and polar-question ($P_{OL}Q$) complements, but not with variants of alternative-question ($A_{LT}Q$) complements or constituent *wh*-complements (Uegaki, 2021, pp. 7–11). Over the years, a number of research have been carried out in this field on the basis of Indo-European languages. Some lean more focus on the semantic interpretation (Anand & Hacquard, 2013; Fischer, 2005), and others put more efforts in accounting for the selectional restrictions (Biezma, 2009; Biezma & Rawlins, 2012; Karttunen, 1977a, 1977b; Pruitt & Roelofsen, 2011; Roelofsen & Farkas, 2015). More recently, combining insights from both lines of research, Uegaki (2021) provides a comprehensive proposal for the lexical entry of *doubt* in English, linking its lexical semantics and selectional restrictions to give a meaning-driven explanation for the latter.

If we zoom in to the dubitative predicate 怀疑(huái yí) in Mandarin Chinese, most attention has been drawn to its well-known division in lexical semantic interpretation when embedded by clausal complements. In such scenarios, the predicate encompasses two interpretations akin to *doubt* (marked as *huaiyi*$_1$) and *suspect* (marked as *huaiyi*$_2$) in English, as revealed in following examples[1]:

(1) a. 我 [$_F$很怀疑] 明天 天气 会 变好。
I very *huaiyi*$_1$ tomorrow weather will become better
"I really doubt that the wether would get better tomorrow."

b. 结合 目前 的 情况， 我怀疑 [他说的是实话]$_{BF}$。
Combine current MOD situation, I *huaiyi*$_2$ he said is truth
"Taking the current situation into consideration, I suspect that what he said is true."

c. 我 怀疑 小唐 [$_F$是不是] 喜欢小强。
I *huaiyi*$_1$ Tang be-NEG-be like Qiang
"I doubt that Tang likes Qiang."

d. 我 怀疑 小唐 是不是 [喜欢小强]$_{BF}$。
I *huaiyi*$_2$ Tang be-NEG-be like Qiang
"I suspect that Tang likes Qiang."

Examples above include both declarative and B(e)-not-B(e) interrogative (as a variant of A-not-A questions discussed in Hagstrom (2006)) clausal complements. Various research have made efforts to identify the strategy to disambiguate these two interpretations from perspectives of grammatical object of the complement (Li, 1987), the descriptive phonological focus distribution (Han, 2001), the "ownership" relationship between the agent and the proposition in the complement (Zhang & Wu, 2013), the pragmatic emotive evaluation of the complement (Yuan, 2014) and the influence of immediate contextual evidence (ICE) (Lu, 2016). Additionally, there are also a few discussions regarding the compositional process in "怀疑(huái yí) + B(e)-not-B(e) interrogative complement" (Li, 1987; Yuan, 2014; Zhang & Wu, 2013). In 2020, my thesis (Huang, 2020) proposed a semantic framework for 怀疑(huái yí) with a lexical entry inspired by Anand and Hacquard (2013) and a disambiguation mechanism based on the information structure embodied by phonological focus which can be formalized in focus semantic value (Rooth,

[1] The subscript F stands for narrow phonological focus, while BF stands for broad phonological focus.

1992, 2016, 1985) based on insights from research aforementioned. I also analyzed that B(e)-not-B(e) interrogative complements are similar to *whether*-complements in English and should be categorized as $P_{OL}Q$. Therefore, the chosen entry of 怀疑(huái yí) is combined with the "highlighted" proposition as proposed in Pruitt and Roelofsen (2011) and Roelofsen and Farkas (2015) to form the meaning of the whole sentence. This analysis aligns with observations made for *doubt whether* sentences in literature and can serve as a cross-linguistic evidence. However, the thesis didn't go further into discussions about other selectional restrictions, and it didn't dive into the potential complication of 怀疑(huái yí) under negation.

This paper is mainly with following tasks: 1) review the classical pieces of research on dubitative predicates that laid motivations and foundations for a more comprehensive semantic analysis (section 2); 2) review results of the recent comprehensive analysis proposed in Uegaki (2021) with analysis with a fashion leaning more to Inquisitive Semantics (section 3); 3) in the light of my previous research in Mandarin 怀疑(huái yí), take a small step forward to extend the comprehensive semantic analysis for English *doubt* to Mandarin 怀疑(huái yí) and verify if it will also yield convincing semantic interpretation and desirable results regarding selectional restrictions (section 4).

2 Classical Research on Dubitative Predicates

2.1 Lexical semantic interpretations of dubitative verbs

Intuitively, dubitative predicates represent doxastic attitudes of an agent that can be placed somewhere in between on the scale from "disbelieve" to "believe". Cross-linguistic counterparts similar to *doubt* is leaning more towards the "disbelieve" side but not to the extent of total disbelief, while the ones similar to *suspect* seem to be closer to the "believe" side but are not to the extent of total belief. Here I will briefly introduce two pieces of research aiming to turn this intuition into formal lexical entries.

2.1.1 Fischer (2005)

Fischer (2005) provides a probabilistic representation of this intuitive reading regarding the interpretation of German dubitative verbs *zweifeln* (similar to *doubt*) and *vermuten* (similar to *suspect*), in terms of the agent's subjective probability (denoted as $P(a)(p)$). The proposal for *zweifeln* embedded by declaratives, $P_{OL}Q$'s and $A_{LT}Q$'s is exemplifeid as follows:

(2) a. $[\![$Maria zweifelt dass p "Maria doubts that p"$]\!] \Leftrightarrow P(m)(p) < 1$
b. $[\![$Maria zweifelt ob p "Maria doubts whether p"$]\!] \Leftrightarrow P(m)(p) < 1\&\ P(m)(\neg p) < 1$
c. $[\![$Maria zweifelt ob p oder q "Maria doubts whether p or q"$]\!] \Leftrightarrow P(m)(p) < 1\&\ P(m)(q) < 1$

One desirable result is that the fact of the negation of *zweifelt dass* (*doubt that*) implying belief can be accounted for by (2-a), as applying negation to this entry will yield $P(m)(p) = 1$. Additionally, he proposes a similar entry for *vermuten* (similar to *suspect*):

(3) $[\![x$ vermuten dass p "x suspects that p"$]\!] \Leftrightarrow 0.5 \leq P(x)(p) < 1$

Combining (2) and (3), he claims that sentences listed are in a construction-level competition and this leads to a scalar implicature of the subjective probability between *zweifelt dass*, *zweifelt ob* and *vermuten dass*. As a result, the interpretation of $[\![$Maria zweifelt dass p "Maria doubts that p"$]\!]$ can be pragmatically "strengthened" to $P(m)(p) < 0.5$ and $[\![$Maria zweifelt ob p "Maria doubts whether p"$]\!]$ leads to the interpretation $0 < P(m)(p) < 0.5$.

However, this analysis is faced with a methodological problem. The scalar implicature analysis based on the construction-level competetion is not gaining empirical support from other predicates. As observed by Uegaki (2021), one can also assume this competition between *be certain whether p* and *believe that p*. With this assumption, the former would implicate $\neg$(*believe p*), while this is equivalent to *believe* $\neg p$ and seems to be an incorrect prediction for the meaning of *be certain whether p*.

2.1.2 Anand and Hacquard (2013)

Another classic research is Anand and Hacquard (2013) which is on the basis of a famous division of attitudal predicates proposed by Bolinger (1968): the first category is representational ones conveying a mental picture and quantifying over information states (e.g. *believe*, *know*); while the other category is non-representational ones that carry out preferential comparisons between alternatives involved in the complements without interacting with information states (e.g. *want*). Anand and Hacquard (2013) observes that emotive doxastics like *hope* and dubitatives like *doubt* don't fall clearly into the two categories, and the claim is that they are with "hybrid semantics" involving both existential quantification over doxastic state (the doxastic assertion) and preferential comparisons over alternatives (the preferential assertion)[2]. For the specific "hybrid semantics" of *doubt* in English, they provide the following entry (Anand & Hacquard, 2013, p. 36):

(4) $[\![\text{a doubts}_C \text{ that } \varphi\ ]\!]^{c,w,S,g}$ is defined iff
φ-verifiers in $S' \neq \emptyset \wedge$ φ-falsifiers in $S' \neq \emptyset$. **[Uncertainty Condition]**
If defined = 1 iff
$\exists w' \in S' : \left[[\![\varphi]\!]^{c,w,S,g} = 1\right] \wedge$ **[Doxastic Assertion]**
p-falsifiers $>_{PROB_{a,w}}$ p-verifiers **[Preference Assertion]**
where $S' = DOX_{a,w}$ and φ-verifiers in S' =
$\lambda S''.S'' \subset S' \wedge \forall S''' \subset S'' : \left[\forall w' \in S''' : [[\![p]\!]^{c,w',S''',g} = 1]\right]$ $(= pow(S' \cap p))$
φ-falsifiers in $S' = \neg\varphi$-verifiers in S'

This entry starts by expressing an uncertainty in the doxastic state such that neither the set of worlds verifying the complement containing p (φ-verifiers) nor the set of worlds falsifying it (φ-falsifiers) in the subject's doxastic state should be empty. This grasps the intuition that *doubt* is rather a "tentative" doxastic attitude without totally negating either of the possibilities. This is named as "the uncertainty condition". Then there is an assertion of doxastic possibility, represented by an existential quantifier over the doxastic information state of the subject, requiring some of them to verify p, namely the content proposition in the complement. This is also in accordance with the uncertainty condition. Last but not least, when interpreting *doubt*, there is a tendency for the subject to take the p-falsifiers as more probable to hold than the p-verifiers, which is named 'the preference assertion'.

This proposal gives a rather satisfactory formal account for the lexical semantics of *doubt* in English, grasping important intuitions we have for its meaning. But a pity is that when put under negation, the entry doesn't yield the desired meaning of *believe p*, as it's not equivalent to "a doesn't consider not p more likely than p". Apart from this, this account doesn't provide much potential for investigations into the selectional restrictions of the predicate, although this is not the main goal of this research.

2.2 Selectional restrictions of dubitative verbs

A large portion of research on this line can trace back to observations made in (Karttunen, 1977a, 1977b) about *doubt* in English, summarized into the following examples:

(5) a. I doubt that they serve breakfast. (Declarative)
b. I doubt whether they serve breakfast. $(P_{OL}Q)$
c. I doubt whether they serve tea-or-coffee$_{\downarrow}$. $(P_{OL}Q$, "fake" $A_{LT}Q)$
d. *I doubt whether they serve tea$^{\uparrow}$ or coffee$_{\downarrow}$. $(A_{LT}Q)$
e. *I doubt whether they serve breakfast or not. $(A_{LT}Q_V N)$
f. *I doubt who serve the breakfast. (Constituent WH)

Unlike most other clause-embedding predicates in English showing a conformity in their compatibility of interrogative complements (i.e. they are either rogative, anti-rogative or responsive), *doubt* shows an inner variation[3] in accepting $P_{OL}Q$ but no other types of interrogatives. It's worth noting that complements of (5-c) and (5-d) are identical on the first sight, but they belong to $P_{OL}Q$ and $A_{LT}Q$ respectively. This can be revealed with insights from Ciardelli et al. (2019a) regarding their intonation structures: in the complement of (5-c), disjuncts of "tea" and

[2] A more recent research by Uegaki and Sudo (2019, p. 337) proposes an alternative formal expression to represent this doxastic assertion as "the comparison class C is restricted to those alternatives that are compatible with the subject's belief", instead of directly quantifying over the subject's doxastic state. This is helpful in differentiating emotive factives and non-veridical emotive doxasitics.

[3] Another predicate that shows this inner variation among interrogative complements is the emotive factive *surprise*, which is with a variation in accepting constituent *wh-* complements but no other interrogatives. But it's clear that its point of variation differ from *doubt*.

"coffee" are pronounced within a single intonational phrase (represented by hyphens), thus it should be analyzed as a polar disjunctive question with a single-clause list structure, whose resolution requires only establishing "they serve either tea or coffee" or "the serve neither tea nor coffee"; while the complement of (5-d) is an alternative question with multi-clausal list structure brought by different intonations on "tea" and "coffee", which presupposes "they serve either tea or coffee" and raises an issue whose resolution requires establishing which specific drink they serve. As for the complement type in (5-e), Biezma (2009) and Biezma and Rawlins (2012) analyze them as a special kind of alternative questions $A_{LT}Q_{V}N$ formed by two opposite alternatives p and $\neg p$.

2.2.1 The traditional views

Karttunen (1977a, 1977b) approached the explanation of this inner variation based on his observation such that (5-a) and (5-b) are equivalent in that they can both be interpreted as "I don't quite believe that they serve breakfast". Therefore, he claims that *whether*-clauses without multi-clause disjucntions are not "genuine embedded questions" that can be represented as sets of various propositions, from which other question-embedding (i.e. rogative or responsive) predicates choose the one that is compatible with the evaluation world to combine legitimately. Instead, they are analyzed as single content propositions (similar to declarative complements). In such a way, the observed inner variation is wiped out and *doubt* can be neatly put into the category of predicates that is not question-embedding (i.e. anti-rogative).

Another research assuming the equivalence between sentences like (5-a) and (5-b) is Biezma and Rawlins (2012). Undertaking the method of Alternative Semantics, where denotations of declarative statements and inquisitive ones are uniformly treated as sets of propositions (Hamblin, 1973), they further point out that the semantic denotation of $P_{OL}Q$'s are singleton proposition sets, which are equivalent to the semantic denotation of declarative statements. It was only after an anti-singleton coercion brought by the pragmatic non-singleton presupposition [4]are sets of bipolar alternative propositions derived from $P_{OL}Q$'s. Meanwhile, other types of interrogatives are analyzed to possess multi-member-set denotations and don't need to go through this anti-singleton coercion. On the other hand, they propose that *doubt* "S-selects for singleton alternative sets" (Biezma & Rawlins, 2012, p. 395) as a restriction on its complements, accounting for its compatibility with declarative and $P_{OL}Q$ complements and its incompatibility with other types of interrogative complements.

It can be noticed that this line of analysis propose a differentiation of essential semantic denotation of $P_{OL}Q$'s from other types of interrogatives and the similarity between $P_{OL}Q$'s and declarative statements, but they are not very intuitive in our natural language perception, as the interrogative perception from $P_{OL}Q$'s are rather robust and no weaker than other question types (especially when in the form of matrix question). Furthermore, their common assumption of the equivalence between "*doubt whether*" and "*doubt that*" is challenged by observations made in Dixon (2005, p. 239):

(6) a. A: Do they serve scrambled eggs?
B: I doubt that / whether they serve scrambled eggs.
b. A: (I bet / assume / am sure) they serve scrambled eggs.
B: I doubt that / # whether they serve scrambled eggs.
c. I don't doubt that he is sick.
$\rightsquigarrow$ "I believe that he is sick."
d. ?? I don't doubt whether he is sick.

It's clear that in (6-a) and (6-b), with different contexts actualized by utterances from the speaker A, the compatibility of *doubt that* and *doubt whether* becomes varied, revealing their non-equivalence. Additionally, this non-equivalence can also be perceived from different compatibility conditions between *doubt that* and *doubt whether* under the same negative environment.

2.2.2 Insights from Inquisitive Semantics

On the other hand, the framework of Inquisitive Semantics $InqB$ and its extension $InqB^{\pm}$ proposed in Ciardelli et al. (2019b), Pruitt and Roelofsen (2011), and Roelofsen and Farkas (2015) seems to be coping better with the two issues pointed out above. Firstly, $InqB$ doesn't presuppose essential difference in nature between the denotation (i.e. the ordinary semantic value, "o-value") of $P_{OL}Q$'s and other types of interrogatives. The framework regards

[4]Please refer to Biezma and Rawlins (2012, p. 393) for technical details

propositions expressed by all sentences as pairs of $\langle \alpha, I \rangle$, in which α stands for an information state (set of possible worlds) capturing the informative content, and I stands for an issue (set of information states, equivalent to the proposition itself) grasping the inquisitive content, as defined in Ciardelli et al. (2019b). This leads to a uniform formal treatment of propositions, no matter declarative or interrogative, as downward-closed sets of information states. The o-value of an inquisitive proposition is with multiple alternatives (i.e. maximal elements), while the o-value of a purely informative (declarative) proposition is with only one alternative. General definitions for o-values of different type of complements are given as follows, accompanied by examples we mentioned in previous sections:

(7) a. **Declarative**
$[\![\text{that } p]\!]^o = \{\{w\text{: } p \text{ is true in } w\}\}^{\downarrow}$
$[\![\text{that they serve scrambled eggs}]\!]^o = \{\{w\text{: they serve scrambled eggs in } w\}\}^{\downarrow}$

b. $\mathbf{P_{OL}Q}$
$[\![\text{whether } p]\!]^o = \{\{w\text{: } p \text{ is true in } w\}, \{w\text{: } p \text{ is false in } w\}\}^{\downarrow}$
$[\![\text{whether they serve scrambled eggs}]\!]^o = \{\{w\text{: they serve scrambled eggs in } w\}, \{w\text{: they don't serve scrambled eggs in } w\}\}^{\downarrow}$

c. $\mathbf{A_{LT}Q_VN}$
$[\![\text{whether } p \text{ or not}]\!]^o = \{\{w\text{: } p \text{ is true in } w\}, \{w\text{: } p \text{ is false in } w\}\}^{\downarrow}$
$[\![\text{whether they serve scrambled eggs or not}]\!]^o = \{\{w\text{: they serve scrambled eggs in } w\}, \{w\text{: they don't serve scrambled eggs in } w\}\}^{\downarrow}$

d. $\mathbf{A_{LT}Q}$
$[\![\text{whether } p^{\uparrow} \text{ or } q_{\downarrow}]\!]^o = \{\{w\text{: } p \text{ is true in } w\}, \{w\text{: } q \text{ is true in } w\}\}^{\downarrow}$
$[\![\text{whether they serve tea}^{\uparrow} \text{ or coffee}_{\downarrow}]\!]^o = \{\{w\text{: they serve tea in } w\}, \{w\text{: they serve coffee in } w\}\}^{\downarrow}$

e. **Constituent WH, mention-all reading**[5]
$[\![\text{wh- } P]\!]^o = \{s |\ \forall d \in D : s \subseteq \{w\text{: } P_d \text{ is true in } w\} \text{ or } s \subseteq \{w\text{: } P_d \text{ is false in } w\}\}$
$[\![\text{who serve breakfast}]\!]^o =$
$\{s |\ \forall d \in D : s \subseteq \{w\text{: d serves breakfast in } w\} \text{ or } s \subseteq \{w\text{: d doesn't serve breakfast in } w\}\}$

f. **Constituent WH, mention-some reading**[6]
$[\![\text{wh-}P]\!]^o = \{\{w\text{: } P_d \text{ is true in } w\} \mid d \in D\}^{\downarrow}$
$[\![\text{who serve breakfast}]\!]^o = \{\{w\text{: d serves breakfast in } w\} \mid d \in D\}^{\downarrow}$

Secondly, their $InqB^{\pm}$ extension proposes a "highlighting" semantic value (h-value) of propositions (marked as $[\![\varphi]\!]^{\pm}$ or $[\![\varphi]\!]^h$) parallel to the o-value we discussed above. The main motivation is to determine discourse referents and provide a toolkit to account for different compatibility conditions of anaphoric discoursal expressions to different types of interrogatives. But meanwhile, insights are also drawn in explaining a peculiarity of *doubt whether* sentences. The idea of the h-value is to track "possibilities that are made particularly salient when (a proposition) φ is uttered and thereby come to serve as highly accessible potential antecedents for subsequent anaphoric expressions" (Roelofsen & Farkas, 2015, p. 378). Through a series of probing, it can be determined that anaphoric discoursal expressions like "so" and "yes / no" responses are sensitive to them, such that: 1) they are only felicitous when the highlighted semantic value is the singleton set of a unique possibility (information state); 2) if 1) is the case, they are referring to this unique possibility. It can be observed that these anaphoric discoursal expressions are felicitous in responses to declaratives and $P_{OL}Q$'s that are with a singleton set as their h-values, but they are not felicitous in responses to $A_{LT}Q$'s and $A_{LT}Q_VN$'s that are with non-singleton sets as h-values. [7] As for constituent *wh*-interrogatives, Roelofsen and Farkas (2015) follows the idea from Groenendijk and Stokhof (1984) in partition semantics to regard their h-values as "abstracts", which are n-place (n is determined by the number of *wh*-elements in the question) properties that are made salient by uttering them. We give general definitions of h-values of different types of complements accompanied by examples from previous sections as follows:

(8) a. **Declarative**
$[\![\text{that } p]\!]^h = \{\{w\text{: } p \text{ in } w\}\}$
$[\![\text{that they serve scrambled eggs}]\!]^h = \{\{w\text{: they serve scrambled eggs in } w\}\}$

b. $\mathbf{P_{OL}Q}$

[5]Adopted from Ciardelli et al. (2019c), s stands for information states; d stands for individual entities in the domain D.

[6]Adopted from Ciardelli et al. (2019c), d stands for individual entities in the domain D.

[7]It's worth noting in $InqB$ representations, $A_{LT}Q$'s $A_{LT}Q_VN$'s are both with $\vee$ as the highest-scoping operator(i.e. $p \vee q$, $p \vee \neg p$ respectively), and propositions with this trait are all with non-singleton h-value; while $P_{OL}Q$'s belong to the category with projection operator as the highest-scoping operator, where propositions are always with singleton h-value. Please refer to the original recursive definitions of h-value in Roelofsen and Farkas (2015, p. 380) for details.

$[\![\text{whether } p]\!]^h = \{\{w\text{: } p \text{ is true in } w\}\}$
$[\![\text{whether they serve scrambled eggs}]\!]^h = \{\{w\text{: they serve scrambled eggs in } w\}\}$

c. **$\mathbf{A_{LT}Q_{V}N}$**
$[\![\text{whether } p \text{ or not}]\!]^h = \{\{w\text{: } p \text{ is true in } w\}, \{w\text{: } p \text{ is false in } w\}\}$
$[\![\text{whether they serve scrambled eggs or not}]\!]^h = \{\{w\text{: they serve scrambled eggs in } w\}, \{w\text{: they don't serve scrambled eggs in } w\}\}$

d. **$\mathbf{A_{LT}Q}$**
$[\![\text{whether } p^{\uparrow} \text{ or } q_{\downarrow}]\!]^h = \{\{w\text{: } p \text{ is true in } w\}, \{w\text{: } q \text{ is true in } w\}\}$
$[\![\text{whether they serve tea}^{\uparrow} \text{ or coffee}_{\downarrow}]\!]^h = \{\{w\text{: they serve tea in } w\}, \{w\text{: they serve coffee in } w\}\}$

e. **Constituent WH**
$[\![\text{wh- } P]\!]^h = \{\lambda x \lambda w[P_w(x)]\}$
$[\![\text{who serve breakfast}]\!]^h = \{\lambda x \lambda w[\text{serve-breakfast}_w(x)]\}$

Therefore, with different types of h-values observed above overlapping with compatibility distributions with *doubt* discussed in (5), it's possible to give an account from this aspect instead of stipulating the essential semantic denotations of complements. Other than this, the following examples taken from Pruitt and Roelofsen (2011, p. 27) also provide evidence supporting that h-value plays a role in the interpretation of *doubt whether* sentences:

(9) a. John doubts whether the door is open.
b. John doubts whether the door is closed.

There is a strong intuition such that (9-a) and (9-b) are not truth-conditionally equivalent. While this cannot be explained from the perspective of essential semantic denotation of complements, as both denote the same set $\{\{w\text{: the door is open in } w\}, \{w\text{: the door is closed in } w\}\}^{\downarrow}$, their h-values differ based on their surface forms that are in contrast, corresponding to one of the alternative possibilities in the o-value and aligning with our intuitions. Last but not least, the different essential semantic denotations of declaratives and $P_{OL}Q$'s in the framework also provide a potential way out for the observations made in (6).

To summarize, proposals made in the framework of inquisitive semantics discussed above seem to possess a good potential to be combined with an appropriate lexical entry of dubitative predicates in accounting for their selectional restrictions.

3 A comprehensive semantic analysis for *doubt* in English

Having reviewed the important literature that made clear what a semantic framework for dubitative predicates should account for in both lexical semantic interpretation and selectional restrictions, together with technical foundations made available in these research, we now turn to Uegaki (2021) and introduce his comprehensive proposal for *doubt* in English and review its predictive. The proposed lexical entry is given as follows:

(10) **The Lexical Entry for *doubt***

- $[\![x \text{ doubts } \varphi]\!]^o(w)$ presupposes:
  (i) $\exists Q \in \mathit{table} : [\![\varphi]\!]^o \subseteq Q$; and
  (ii) $\forall p \in [\![\varphi]\!]^o : DOX_x^w \cap p \neq \emptyset$ **[Presuppositional components]**
- If the above presuppositions are met,
  $[\![x \text{ doubts } \varphi]\!]^o(w) = 1 \text{ iff } DOX_x^w \nsubseteq \bigcup [\![\varphi]\!]^h$ **[Assertive component]**
- The exhaustification operator and the pruning mechanism is at work to strengthen the assertive component.

I will briefly review this proposal's predictions in key issues we've discussed in previous sections from both perspectives of lexical semantic interpretation and selectional restrictions.

3.1 Predictions regarding lexical semantic interpretation

(I) **The interpretation in affirmative contexts**
The presupposition (i) requires that the o-value of the complement should be the subset of some proposition that is in the conversation table i.e. the set of propositions made so far in the conversation either by statements

asserted or questions raised by conversation participants. The presupposition (ii) in the entry requires that for every set of possible worlds included in the o-value (as illustrated in (7), thereby supporting an alternative of the proposition's issue) of the complement, its intersection with the subject's doxastic state should not be empty. This accurately grasps the intuition that the doxastic attitude conveyed by *doubt* is "uncertain" and weaker than "disbelieve", such that the subject's doxastic state could not completely rule out any alternative in the issue embodied by the proposition in the complement. This aligns with the uncertainty condition of Anand and Hacquard (2013) proposal. On the other hand, the assertive component requires that possible worlds in the subject's doxastic state are not all elements of information state(s) in the complement's h-value. In other words, this requires only an existential quantification over the subject's doxastic state to be compatible with any information state in complement's h-value. This, on the other hand, captures the existential doxastic attitude towards the negation of complement's h-value, which aligns with the doxastic assertion in Anand and Hacquard (2013).

(II) **The interpretation in negative contexts**
The aim is to get the "*believe*" interpretation when the negation of *doubt* is embedded with an arbitrary declarative complement (i.e. in the form of "x doubts that p"). The assertive component can predict exactly this, as it will yield $DOX_x^w \subseteq \bigcup [\![\varphi]\!]^h$. In the scenario of "$x$ doubts that p", $\bigcup [\![\varphi]\!]^h$ is equivalent to the single alternative of p i.e. the informative content of p. This represents that all worlds in the subject's doxastic state is compatible to the inforamtive content of p, conveying a doxastic attitude equivalent to a belief in p, as shown in (6-c).

(III) **The preferential comparison**
It can be noticed that there seems to be a trade-off relation between the account for the interpretation in negative contexts discussed above and the preferential comparison (i.e. the unequal probabilistic perception of alternatives) as pointed out by the preference assertion in Anand and Hacquard (2013). In the entry of Uegaki (2021), the assertive component captures the former but is not capable for the latter, being contrary to the Anand and Hacquard (2013). Additionaly, the assertive component can be directly entailed by presupposition (ii) when the o-value of the complement contains two alternatives in contrast. For the solution of this dilemma, the exhaustification operator and a pragmatic pruning mechanism proposed by Bar-Lev (2020) and Bar-Lev and Fox (2020) are introduced. The claim is that for a modal / attitudal statement, when there is not a scalemate with a stronger modal force, the statement can be "strengthened" through this exhaustification operator to fill in the gap. Here for *doubt* in English, this operator is claimed to be at work, as potential candidates for its stronger scalemate *disbelieve* and *disagree* are with more complex morphology and different argument structures, this scalemate is analyzed as absent. But meanwhile, the doxastic attitude conveyed by *doubt* is not strong as "disbelieve", and it's claimed that the pruning mechanism plays a role in this to reduce the modal force to the extent that is plausible in the context. As a final result, the actual modal force is in the range that is stronger than the existential statement in the assertive component but also weaker than the necessity statement leading to the "disbelief" interpretation. With this method, both the interpretation under negation and the preferential comparison are grasped. On the other hand, with the general assumption such that exhaustification only strengthens the overall meaning and is not applied to downward-entailing context (Chierchia et al., 2013), the interpretation in negative contexts are not affected and the deriving process discussed in (II) is maintained.

3.2 Predictions regarding selectional restrictions

(I) **The compatibility with $P_{OL}Q$ and the observed similarity with declarative complements**
Suppose p is an arbitrary sentence, then by (7) we have the declarative complement $[\![\text{that } p]\!]^o = \{\{w : p \text{ is true in } w\}\}^{\downarrow}$ and the $P_{OL}Q$ complement $[\![\text{whether } p]\!]^o = \{\{w : p \text{ is true in } w\}, \{w : p \text{ is false in } w\}\}^{\downarrow}$. Then we apply them respectively to the proposed entry for *doubt*:

(11) a. •$[\![x \text{ doubts that } p]\!]^o(w)$ presupposes:
(i)$\exists Q \in table : \{\{w : p \text{ is true in } w\}\}^{\downarrow} \subseteq Q$; and
(ii)$\forall p' \in \{\{w : p \text{ is true in } w\}\}^{\downarrow} : DOX_x^w \cap p' \neq \emptyset$
•If the above presuppositions are met,
$[\![x \text{ doubts that } p]\!]^o(w) = 1$ iff $DOX_x^w \nsubseteq \{w : p \text{ is true in } w\}$

b. •$[\![x \text{ doubts whether } p]\!]^o(w)$ presupposes:

(i)$\exists Q \in table : \{\{w : \ p \text{ is true in } w\}, \{w : \ p \text{ is false in } w\}\}^{\downarrow} \subseteq Q$; and
(ii)$\forall p' \in \{\{w : \ p \text{ is true in } w\}, \{w : \ p \text{ is false in } w\}\}^{\downarrow} : DOX_x^w \cap p' \neq \emptyset$

•If the above presuppositions are met,
$[\![x \text{ doubts whether } p]\!]^o(w) = 1$ iff $DOX_x^w \nsubseteq \{w : p \text{ is true in } w\}$

There is no contradiction or triviality found in the applications above, aligning with the predicate's compatibility with both declarative and $P_{OL}Q$ complements. Another important observation is that in both applications, results yielded from the assertive component are identical, as it is only sensitive to h-value of the complement. This can serve as a reasonable explanation for the similarity between these two types of complements from which Biezma and Rawlins (2012) and Karttunen (1977a, 1977b) assumed their equivalence, as we mentioned in 2.2.1.

(II) **The observed difference between $P_{OL}Q$ and declarative complements**
In 2.2.1, it is mentioned that (6-a) and (6-b) taken from Dixon (2005) can serve as counterexamples to the assumption that declarative and $P_{OL}Q$ complements should be regarded as identical when embedded to *doubt*, as they have different compatibility conditions in certain contexts. In the current framework, this can be accounted for with respect to the satisfaction condition of presupposition (i). In(6-a) and (6-b), different propositions are brought into the conversation table by different utterances from A. In (6-a), the conversation table contains the inquisitive proposition "Do they serve scrambled eggs?". If we label "they serve scrambled eggs" as p, then this inquisitive proposition can be formally represented as $\{\{w : p \text{ is true in } w\}, \{w : p \text{ is false in } w\}\}^{\downarrow}$. Then from what we laid out in (11-a) and (11-b), we can see that o-values for both *that p* and *whether p* are subsets of this inquisitive proposition raised. Therefore, presupposition (i) is satisfied in both cases, aligning with the fact that *doubt that* and *doubt whether* are both acceptable in B's response in (6-a). However, in (6-b), the conversation table contains an informative proposition "they serve scrambled eggs" in the form of $\{\{w : p \text{ is true in } w\}\}^{\downarrow}$. In this case, only the set represented by *that p* is a subset of this asserted proposition, aligning with the fact that only *doubt that* is acceptable in B's response in (6-b). This analysis serves as a support for the claim that $P_{OL}Q$'s are still essentially different from declaratives, and as a result, their licensing conditions under *doubt* differ in various contexts.

(III) **Incompatibilities with $A_{LT}Q$, $A_{LT}Q_VN$ and constituent *wh*-complements**
In the current framework, this restriction can be accounted for with respect to h-values of these types of complements. Recall that the assertive component of the entry requires $DOX_x^w \nsubseteq \bigcup[\![\varphi]\!]^h$. Then let's examine the three types of interrogative complements: 1) With an $A_{LT}Q$ complement in the form of *whether* $p^{\uparrow}$ *or* $q_{\downarrow}$, $\bigcup[\![\text{whether } p^{\uparrow} \text{ or } q_{\downarrow}]\!]^h = \{w : p \text{ is true in } w\} \cup \{w : q \text{ is true in } w\}$. Since this $A_{LT}Q$ is a closed interrogative, the possibility of "neither p nor q is true" has been ruled out through presupposition from the subject's doxastic state as proposed in Ciardelli et al. (2019c). However, the assertive component here requires $DOX_x^w \nsubseteq \{w : p \text{ is true in } w\} \cup \{w : q \text{ is true in } w\}$ i.e. the subject's doxastic state to be compatible with some possibility where "neither p nor q is true" is the case, causing a contradiction; 2) With an $A_{LT}Q_VN$ complement in the form of *whether p or not*, $\bigcup[\![\text{whether } p \text{ or not}]\!]^h = \{w : p \text{ is true in } w\} \cup \{w : p \text{ is false in } w\}$. Here, the disjunction is the trivial information stage covering the whole logical space W and this leads to $DOX_x^w \nsubseteq W$, which is a contradiction; 3) With a constituent *wh*-complement, any member of its h-value is of type $\langle e, st\rangle$, which is different from the the type of doxastic states (as information states) $\langle s, t\rangle$. As a result, the assertive component will be trivially true due to this type clash, as the subset relation never holds between objects with different types.

Therefore, we can conclude that these three types of interrogative complements will cause systematic contradiction or triviality under the current entry for *doubt*, aligning with incompatibilities in empirical linguistic observations.

(IV) **The incompatibility with $P_{OL}Q$ complement in negative contexts**
For this incompatibility, the current framework accounts for it with a conflict between the presupposition (ii) and the negation of the assertion. More specifically, for *x doesn't doubt whether p*, the presupposition (ii) requires $\forall p' \in \{\{w : p \text{ is true in } w\}, \{w : p \text{ is false in } w\}\}^{\downarrow} : DOX_x^w \cap p' \neq \emptyset$, while the negative assertion requires $DOX_x^w \subseteq \{w : p \text{ is true in } w\}$, which means for information states equal to or stronger than the alternative $\{w : p \text{ is false in } w\}$ of $[\![\text{whether } p]\!]$, their intersections with DOX_x^w are empty. This contradicts with the universal statement embodied by the presupposition (ii), causing the low acceptability of (6-d).

This is in contrast with (6-c) where the complement is declarative, serving as another evidence of the difference between declarative and $P_{OL}Q$ complements. But here the reasoning relies on the assumption that presupposition(ii) in the entry remain identical under negation, which is a little counter-intuitive for me and potentially needs an explicit illustration.

4 Extending to the semantic analysis of 怀疑(huái yí)

4.1 The previous framework for "怀疑"(huái yí)

This section will briefly introduce my previous formal framework proposed for 怀疑(huái yí) in Huang (2020) to serve as a reference of comparison.

As mentioned in the introduction of this paper, most research related to 怀疑(huái yí) in Mandarin focus on the disambiguation strategy between $huaiyi_1$ (similar to *doubt*) and $huaiyi_2$ (similar to *suspect*) included in its semantics. It can be observed that most of them are somehow partial or inaccurate when applied to a wider empirical coverage with potential counterexamples found[8]. However, some of the accounts seems interrelated and can be combined for a more comprehensive proposal, providing the investigating direction for Huang (2020). First of all, there is a descriptive phonological focus distribution observed by Han (2001), such that in cases where 怀疑(huái yí) is embedded by a declarative complement, 怀疑(huái yí) is interpreted as $huaiyi_1$ iff there is no phonological focus of any form in the complement, and it is interpreted as $huaiyi_2$ iff otherwise. By introducing the concept of focus semantic value (FSV) introduced by Rooth (1992, 2016, 1985), the former case is equivalent to the formal expression of $|[\![\text{complement}]\!]^F| = 1$, as the FSV of a non-focused complement proposition is a singleton set containing its o-value; while the latter case is equivalent to $|[\![\text{complement}]\!]^F| > 1$ as focus marking introduces multiple focus alternatives to the set. This can be treated as a representation of the information status as discussed in Zhang and Wu (2013), but not from their claim of "ownership relations" that cause over-generalization. Instead, it fits better to signal the differentiation between "new / prominent information " and "old / given information" as claimed in Hinterwimmer (2012) and Schwarzschild (1999)[9]. Thus, another layer of relationship can be set up: if the proposition in the declarative complement is already given previously (revealed as $|[\![\text{complement}]\!]^F| = 1$), it tends to be interpreted as $huaiyi_1$; if the proposition in the declarative complement is new to the context (revealed as $|[\![\text{complement}]\!]^F| > 1$), 怀疑(huái yí) tends to be interpreted as $huaiyi_2$. [10]This relationship is also found to be valid in cases where 怀疑(huái yí)is embedded by B(e)-not-B(e) interrogative complements with empirical evidence[11]

As for the lexical entry of 怀疑(huái yí), the framework inherited the main spirit of the proposal in Anand and Hacquard (2013) with two adjustments: 1) the preference assertion in the general entry is adjusted to be a neutral unequal relationship between the probability perceptions of complement proposition verifers and complement proposition falsifiers, while the specific unequal direction is only decided after going through the disambiguation mechanism mentioned above; 2) the highlighting mechanism proposed by Pruitt and Roelofsen (2011) and Roelofsen and Farkas (2015) is introduced to the entry to account for the compositional process of sentences with 怀疑(huái yí) + B(e)-not-B(e) interrogative complements, as I argued that B(e)-not-B(e) interrogative complements bear syntactic and semantic similarities with *whether...* complements ($P_{OL}Q$) but differ from *whether...or not* complements ($A_{LT}Q_VN$). Combining with the disambiguation mechanism introduced above, the whole lexical entry is given as follows:

[8]Please refer to the second section of Huang (2020) for a detailed literature review.

[9]A more recent parallel research B. Liu and Yuan (2021) has also regarded information structure as an important interpretation strategy for Mandarin 怀疑(huái yí).

[10]Insights from Lu (2016) can serve as a potential account for the root of this mechanism, since the reasoning process supplemented by immediate contextual evidence he proposes aligns with it. Empirically, when the content proposition in the complement is consistent with the reasoning result of the immediate contextual evidence p (i.e. q as in $p \to \diamond q$), the focus tends to lie in the proposition and $huaiyi_2$ is chosen, whereas when the content proposition is in conflict with the reasoning result from p (i.e. $\neg q$), the focus tends to be out of it and $huaiyi_1$ is chosen. Since the logical reasoning behind the dubitative attitude tends to be immediate, if the agent draws a proposition that is consistent with the latest immediate contextual evidence, it tends to be new and "not given", and the motivation bringing it forward is add its probability to the common ground, so the attitude is $huaiyi_2$. In contrast, when the proposition is not consistent with the latest immediate contextual evidence, the necessity of bringing it into the discourse tends to originate from its "givenness" and a motivation to negatively reconsider it, then the attitude is $huaiyi_1$.

[11]Note that in such cases, the focus may lie on the inquisitive operator "B(e)-not-B(e)". Although superficially it's in the complement clause, it's a syntactically raised operator and a focus on it doesn't bring focus alternatives. $|[\![\text{complement}]\!]^F| = 1$ is still the case under this condition.

(12) For the verb *huaiyi* in the Mandarin natural language, define its metalanguage definition as BIASED-BELIEF.

1.**The definition in the metalanguage:**
BIASED-BELIEF(A, p, w) is defined iff
p-verifiers in $DOX_{A,w} \neq \emptyset \wedge p$-falsifiers in $DOX_{A,w} \neq \emptyset$. **[Uncertainty Condition]**
If defined = 1 iff
$\exists w' \in DOX_{A,w} : [p(w') = 1] \wedge$ **[Doxastic Assertion]**
p-falsifiers $\neq_{PROB_{A,w}}$ p-verifiers **[Preference Assertion]**
where p-verifiers in $DOX_{A,w}$ =
$\lambda S''.S'' \subset DOX_{A,w} \wedge \forall S''' \subset S'' : [\forall w' \in S''' : [p(w') = 1]]$ $(= pow(DOX_{A,w} \cap p))$
p-falsifiers in $DOX_{A,w}$ = $\neg p$-verifiers in $DOX_{A,w}$

2.**The semantics of *huaiyi*:**
$[\![\text{A } huaiyi\ \varphi]\!]^w$ = BIASED-BELIEF (A, ιp[HIGHLIGHT$(p)\wedge p \in [\![\varphi]\!]$],w)

3.**The disambiguation mechanism:**
a. When φ is a clausal complement and $|[\![\varphi]\!]^F| = 1$, $[\![huaiyi]\!] = [\![huaiyi_1]\!]$ i.e. $\exists w' \in DOX_{A,w}$: $[p(w) = 1] \wedge$ p-falsifiers $>_{PROB_{a,w}}$ p-verifiers, where $p = \iota p[\text{HIGHLIGHT}(p) \wedge p \in [\![\varphi]\!]]$.
b. When φ is a clausal complement and $|[\![\varphi]\!]^F| > 1$, $[\![huaiyi]\!] = [\![huaiyi_2]\!]$ i.e. $\exists w' \in DOX_{A,w}$: $[p(w) = 1] \wedge$ p-verifiers $>_{PROB_{a,w}}$ p-falsifiers, where $p = \iota p[\text{HIGHLIGHT}(p) \wedge p \in [\![\varphi]\!]]$.

Apart from the innovation of the disambiguation mechanism specific to Mandarin 怀疑(huái yí), the pros and cons of this similar to those of Anand and Hacquard (2013). The good aspect is that intuitive elements in the interpretation of the dubitative are grasped. But on the other hand, although a step forward is taken to explain the clause-embedding process of a certain kind of interrogative, it didn't further explore more general selectional restrictions. Also, the specific way it adopts the highlighting mechanism seems not very suitable for this further exploration, as it may be rather stipulative for this entry to refuse certain types of complements just because it simply "requires" the highlighted possibility from the complement to be unique without a proper logical justification.

4.2 Testing the alternative framework for 怀疑(huái yí)

In this section, an alternative framework for 怀疑(huái yí) based on Uegaki (2021) will be tested to check if it also works well with empirical observations in Mandarin cases. Comparisons with the previous framework will also be made to examine which one works better from certain perspectives. I will dive into entry construction and testing for $huaiyi_1$ and $huaiyi_2$ respectively.

4.2.1 The alternative entry for $huaiyi_1$ and its predictions

The intuitive interpretation for $huaiyi_1$ is largely similar to *doubt* in English, so here we adopt the same entry and briefly go through predictions as we did in section 3:

(13) •$[\![x\ huaiyi_1\ \varphi]\!]^o(w)$ presupposes:
(i)$\exists Q \in table : [\![\varphi]\!]^o \subseteq Q$; and
(ii)$\forall p \in [\![\varphi]\!]^o : DOX_x^w \cap p \neq \emptyset$ **[Presuppositional components]**

•If the above presuppositions are met,
$[\![x\ huaiyi_1\ \varphi]\!]^o(w) = 1$ iff $DOX_x^w \nsubseteq \bigcup [\![\varphi]\!]^h$ **[Assertive component]**

•The exhaustification operator and the pruning mechanism is at work to strengthen the assertive component.

(I) **The interpretation in affirmative contexts**
The presupposition (i) is in a good match with what we proposed for the disambiguation mechanism between two interpretations of $huaiyi_1$ and $huaiyi_2$: $huaiyi_1$ interpretation is chosen when $[\![\varphi]\!]^F = 1$, which reveals that there is no focus in the complement. It can then be speculated that the proposition of the complement is already part of the "topic" brought onto the conversation table, leading to the complement's o-value being a subset of some proposition on the conversation table. Other than this, the presupposition (ii) aligns with the uncertainty condition, and the assertive component aligns with the doxastic assertion respectively in (12), with similar reasons illustrated in 3.1 (I).

(II) **The interpretation in negative contexts**
It can be observed that when the negation of *huaiyi*$_1$ is embedded by declarative complements, it will also yield the meaning equivalent to *believe*. This is obvious from following examples taken from the CCL corpus:

(14) a. 我[$_F$不怀疑] 她 曾经 幸福过。
I NEG *huaiyi*$_1$ she used to be happy PAST
"I don't doubt that she used to be happy."
⇝我 相信 她 曾经 幸福过。
"I believe that she used to be happy."

b. 他 [$_F$不怀疑] 中国 将成为 一个 经济 强国。
He NEG *huaiyi*$_1$ China will become a MOD economic powerful country
"He doesn't doubt that China will become a country with prosperous economy."
⇝他 相信 中国 将成为 一个 经济 强国。
"He believes that China will become a country with prosperous economy."

(Taken from Center for Chinese Linguistics PKU (2023))

Therefore, the entry makes the correct prediction with the same process we discussed in 3.1(II) (i.e. $DOX_x^w \subseteq \bigcup[\![\varphi]\!]^h$ predicts a belief in p), and this is an improvement compared to (12) in the Anand and Hacquard (2013) fashion.

(III) **The preferential comparison**
Similar to the situation in English, *huaiyi*$_1$ implies that the subject takes the possibility of complement falsifier greater than verifier in the doxastic state, and there is no stronger scalemate for *huaiyi*$_1$ in a strict sense (i.e. with the same morphological complexity). The counterpart of *disbelieve* in Mandarin is "不信"(NEG + believe), which is also with a negative prefix. Therefore, the exhaustification operator and the pruning mechanism discussed in 3.1(III) can also work here to guarantee the doxastic attitude to be stronger than a purely existential statement and grasp the idea of preference assertion in (12). Meanwhile, the interpretation in negative contexts is not affected as the exhaustification operator and the pruning mechanism don't apply to downward-entailing contexts.

(IV) **Embedding $P_{OL}Q$ complements and declarative complements**
Similar to *doubt* in English, *huaiyi*$_1$ can embed both $P_{OL}Q$ complements and declarative complements, and the results from these two types of embedding can bear semantic similarities. This can be reflected from following examples:

(15) a. 我 [$_F$很怀疑] 小唐 喜欢小强。
I very *huaiyi*$_1$ Tang like Qiang
"I really doubt that Tang likes Qiang."

b. 我 很怀疑 小唐 [$_F$是不是] 喜欢小强。
I very *huaiyi*$_1$ Tang be-NEG-be like Qiang
"I really doubt that Tang likes Qiang."

The phenomenon can be grasped through applying sentences like examples above to the new entry in (13) respectively, which will yield identical assertive component with a similar process illustrated in 3.2(I). It's worth noting that the previous framework in (12) also captures this by requiring a unique highlighted proposition to be combined with the predicate. Both $P_{OL}Q$ and declarative complements satisfy this condition, and when the $P_{OL}Q$ is a result by projecting a inquisitive operator "B(e)-not-B(e)" to the declarative, their highlighted propositions are identical. The new entry seems to be less stipulative from this aspect, as this semantic similarity is attained from the results after applying the entry, instead of being a result from a predefined restriction.

On the other hand, the insights about the difference between $P_{OL}Q$ complements and declarative complements in English as observed in (6-a) and (6-b) seems not very applicable in similar scenarios in Mandarin Chinese, according to a preliminary interview with a few native speakers. This is exemplified in the following example[12]:

[12]Here we use deictic expressions for complements in B's responses, as repeating the whole content A mentioned in responses sounds awkward in this scenario.

(16) a. A: 这家 酒店 有 免费早餐 吗?
This hotel have free breakfast Q-particle
A: "Dos this hotel provide free breakfast?" $[\![\psi_1]\!]^o = \{|p|, |\overline{p}|\}^{\downarrow}$[13]
B: 我 [$_F$很怀疑] 他们 有 这个 服务。
I very *huaiyi*$_1$ they have this service
B: I really doubt that they have this service. $[\![\varphi_1]\!]^o = \{|p|\}^{\downarrow}$
B': ?我 很怀疑 他们 [$_F$是不是] 有 这个 服务。
I very *huaiyi*$_1$ they be-NEG-be have this service
B': I really doubt that they have this service. $[\![\varphi_2]\!]^o = \{|p|, |\overline{p}|\}^{\downarrow}$

b. A: 这家 酒店 有 免费早餐。
This hotel have free breakfast
A: "This hotel provides free breakfast." $[\![\psi_2]\!]^o = \{|p|\}^{\downarrow}$
B: ?我 [$_F$很怀疑] 他们 有 这个 服务。
I very *huaiyi*$_1$ they have this service
B: I really doubt that they have this service. $[\![\varphi_3]\!]^o = \{|p|\}^{\downarrow}$
B': 我 很怀疑 他们 [$_F$是不是] 有 这个 服务。
I very *huaiyi*$_1$ they be-NEG-be have this service
B': I really doubt that they have this service. $[\![\varphi_4]\!]^o = \{|p|, |\overline{p}|\}^{\downarrow}$

It seems that both contexts in (16-a) and (16-b) reveal the difference between $P_{OL}Q$ complements and declarative complements in Mandarin with varied acceptability when responding to a inquisitive proposition and a declarative proposition respectively. Here it's rather interesting that for responses found by native speakers to be with a lower acceptability, o-values of their complements are equal to that of the proposition brought to the conversation table. Intuitively, this may be originated from the expectation for a response to either lock onto a more fine-grained possibility in the proposition on the conversation table, or take a step back to reconsider the informative content / presupposition in it. With this insight, it may be reasonable to adjust the presupposition (i) in the alternative entry as follows, which would also maintain the presupposition that there is at least some possibility in the complement that has been brought into the context for consideration when *huaiyi*$_1$ interpretation is activated:

(17) $\exists Q \in table(\exists p(p \in [\![\varphi]\!]^o \wedge p \in Q)) \wedge [\![\varphi]\!]^o \neq Q$

However, it should be noticed that this is only based on judgments from a very limited number of native speakers, and judgments for responses in (16) are rather nuanced. Larger-scale interviews are necessary to verify the observations above and determine if the adjustment to presupposition (i) is required.

(V) **Embedding $A_{LT}Q$, $A_{LT}Q_VN$ and constituent *wh*-complements**
Similar to *doubt* in English, *huaiyi*$_1$ cannot embed $A_{LT}Q$, $A_{LT}Q_VN$ and constituent *wh*-complements. This is rather obvious for the former two types, as sentences similar to the following examples are always ungrammatical:

(18) a. 我[$_F$很怀疑] 他 是广东人 -或者- 福建人。 ($P_{OL}Q$)
I very *huaiyi*$_1$ he is Cantonese -or- Fujianese
"I really doubt that he is a Cantonese-or-Fujianese."

b. *我[$_F$很怀疑] 他 是广东人$^{\uparrow}$ 还是福建人$_{\downarrow}$。 ($A_{LT}Q$)
I very *huaiyi*$_1$ he is Cantonese or Fujianese
Intended: I really doubt whether he is a Cantonese or Fujianese.

c. *我[$_F$很怀疑] 他 是广东人 还是不是广东人。 ($A_{LT}Q_{vN}$)
I very *huaiyi*$_1$ he is Cantonese or not a Cantonese
Intended: I really doubt whether he is a Cantonese or not.

With observations above, the same accounting process discussed in 3.2(III) for $A_{LT}Q$ and $A_{LT}Q_{vN}$ complements can be applied here.

[13]Here I use the notation $|.|$ to abbreviate information states/possibilities. More specifically, $|p| = \{w : p \text{ is true in } w\}$, $|\overline{p}| = \{w : p \text{ is false in } w\}$

The situation for $huaiyi_1$ + constituent *wh*-complements is more complicated, as sentences with this combination are generally grammatical:

(19) a. 我[$_F$很怀疑] 什么 理论 能 解释 这个现象。
I very $huaiyi_2$ what theory can explain this phenomenon
\# "I really doubt what theory can explain this phenomenon."
"I really doubt that there is some (any) theory that can explain this phenomenon."

b. 我[$_F$很怀疑] 谁 能 让他 回心转意。
I very $huaiyi_2$ who can let him change mind
\# I really doubt who can change his mind."
"I really doubt that someone (anyone) can change his mind."

c. 我[$_F$很怀疑] 哪里 有 这么好的 条件。
I very $huaiyi_2$ where have such good conditions
\# "I really doubt where is the place that is with good conditions as such."
"I really doubt that there is some (any) place with good conditions as such."

d. 我[$_F$很怀疑] 哪家 公司 会 录用他。
I very $huaiyi_1$ which company would hire him
\#"I really doubt which company would hire him."
"I really doubt that there is some (any) that would hire him.

However, with a closer examination it's clear that there is a strong inclination to interpret these *wh*-complements as existential statements under modal expressions. This inclination has been supported and analyzed by literature as "quexistentials" or "wh-indefinites functioning as Epistemic Indefinites" in Mandarin (Hengeveld et al., 2022; M. Liu & Yang, 2021), as they can function as both question words and existential indefinites without morphological alternation. Here in (19), it's reasonable to analyze that the interrogative readings of *wh*-complements are blocked due to the same triviality caused by the type-mismatch discussed in 3.2(III) for constituent *wh*-complements[14].

(VI) **$P_{OL}Q$ complements in negative contexts**
This is another part where Mandarin empirical observations show a potential difference from English data. The negation of $huaiyi_1$ seems to fuzzily accept B(e)-not-B(e) interrogative complements for some native speakers, although a supplementary statement following it can serve to make it clearer. This can be illustrated with the following example:

(20) 我[$_F$不怀疑] 小唐 是不是 喜欢小强，(我很 确定他 就是 喜欢小强)。
I don't $huaiyi_1$ Tang be-NEG-be like Qiang. (I very sure he does like Qiang)
"I don't doubt whether Tang likes Qiang. (as I'm sure that he does like Qiang.)"

A potential explanation is that the presupposition (ii) in the alternative entry (13) may be cancelled in the negative environment, so that the contradiction between presupposition (ii) and the assertive component doesn't arise, and the assertive component leads to the reading of a belief in the h-value of the complement. Again, this would require larger-scale interviews with native speakers to confirm the linguistic intuitions and determine if the adjustment to presupposition (ii) is required.

4.2.2 The alternative entry for $huaiyi_2$ and its predictions

We can also try to construct an entry for $huaiyi_2$ in the fashion of Uegaki (2021), but as we will observe in testings, it will be more limited in several aspects.

[14] An interesting observation is that for sentences with $huaiyi_1$ + *wh*-complement, it's also possible to lay the focus on the *wh*-indefinite itself and the quexistential reading still arises:

(i) 我很怀疑 [$_F$谁] 能 让他 回心转意。
I very $huaiyi_1$ who can let him change mind
\# I really doubt who can change his mind."
"I really doubt that someone (anyone) can change his mind."

This is in contrast with the QF Bidirectional supported by Hengeveld et al. (2022), claiming that this focus distribution will realize a interrogative reading.

(21) $\bullet[\![x\ huaiyi_2\ \varphi]\!]^o(w)$ presupposes:

(i)$\forall Q \in table(\forall p(p \in [\![\varphi]\!]^o \to p \notin Q)) \vee (\exists Q \in table([\![\varphi]\!]^o \subset Q))$; and

(ii)$\forall p \in [\![\varphi]\!]^o : DOX_x^w \cap p \neq \emptyset$ **[Presuppositional components]**

$\bullet$If the above presuppositions are met,

$[\![x\ huaiyi_2\ \varphi]\!]^o(w) = 1$ iff $DOX_x^w \nsubseteq \bigcup\{\neg\psi|\ \psi \in [\![\varphi]\!]^h\}$ **[Assertive component]**

(I) **The interpretation in affirmative contexts**

The presupposition (i) requires a disjunction to be satisfied: either for any possibility included in the o-value of the complement, it is not included in any proposition that is already on the conversation table; or the o-value of the complement is a true subset of some proposition on the conversation table. This presupposition also aims to align with the disambiguation mechanism stated in the previous framework (12). As stated before, the $huaiyi_2$ interpretation is activated when the focus lies in the complement and the complement is regarded as new/prominent information. This is the case either when no proposition on the conversation table include any possibility in o-value of the complement at all; or when the o-value of the complement **properly** entails some proposition on the conversation table, so that it rules out more information states and can be tagged as new information through focus. The presupposition (ii) still conveys the uncertainty condition in (12). The assertive component is still similar to the doxastic assertion in (12), but the specific condition is adjusted to emphasize the existential doxastic attitude towards the complement's h-value itself.

(II) **The interpretation in negative contexts and the preferential comparison**

When the negation of $huaiyi_2$ is embedded by declarative complements, it will yield the meaning equivalent to *don't think it possible* or *disbelieve*:

(22) 只有 维护了 规则 的尊严， 人们 才 不怀疑 规则 是 人为的。
only maintain PERF rules ' dignity, people only then NEG $huaiyi_2$ rules are imposed
"Only when the diginity of rules are maintained would people stop suspecting that they are purely imposed."
⇝只有 维护了 规则 的 尊严， 人们 才 不认为 规则 是 人为的。
"Only when the diginity of rules are maintained would people stop thinking/believing that they are purely imposed."

(Taken from Center for Chinese Linguistics PKU (2023))

The adjusted assertive component can grasp this interpretation under negation, as the negation of the assertive component regarding a declarative complement would be $DOX_x^w \nsubseteq \bigcup\{\neg\psi|\ \psi \in [\![\varphi]\!]^h\}$. When the complement is a declarative with a purely informative proposition p, $\bigcup\{\neg\psi|\ \psi \in [\![\varphi]\!]^h\}$ is equivalent to the informative content of $\neg p$. Then this represents that all worlds in the subject's doxastic state is compatible to the informative content of $\neg p$, conveying a doxastic attitude equivalent to a belief in $\neg p$ i.e. $\neg$*believe* p. This can be taken as an improvement compared to the previous framework (12).

However, the trade-off effect between grasping the interpretation in negative contexts and successfully representing the preferential comparison arises again, as the exhaustification operator and the pruning mechanism seems not applicable here. This is because 相信("believe") can be analyzed as a scalemate of $huaiyi_2$ with stronger modal force. If we add some form of probabilistic comparative statement (similar to preference assertion in (12)) to form a conjunction with the assertive component in the current proposal, then its negation would be a disjunction that can be satisfied with only the negation of this comparative statement. As a result, the explanation for the negative interpretation would be lost. Therefore, more investigations are needed to come up with another method that can accounting for the interpretation in negative contexts and the preferential comparison at the same time.

(III) **Embedding $P_{OL}Q$ complements and declarative complements**

$huaiyi_2$ can also embed both $P_{OL}Q$ complements and declarative complements, and the results from these two types of embedding can bear semantic similarities, as shown by following examples:

(23) a. 我 怀疑 小唐 [喜欢小强]$_{BF}$。
I $huaiyi_2$ Tang like Qiang
"I suspect that Tang likes Qiang."

b. 我 怀疑 小唐 是不是 [喜欢小强]$_{BF}$。
I $huaiyi_2$ Tang be-NEG-be like Qiang
"I suspect that Tang likes Qiang."

The phenomenon can be grasped through applying sentences like examples above to the current entry, since the assertive component will yield the identical result $DOX_x^w \nsubseteq |\overline{p}|$. This is also captured in (12) with the requirement for a unique highlighted proposition to be combined with the predicate, but less stipulative, similar to what we've discussed in 4.2.1(IV).

The situation regarding the difference between $P_{OL}Q$ complements and declarative complements is also a bit complicated under $huaiyi_2$. The following group of examples serves as an illustration:

(24) a. A: 这家 酒店 有 免费早餐 吗?
This hotel have free breakfast Q-particle
A: "Dos this hotel provide free breakfast?" $[\![\psi_1]\!]^o = \{|p|, |\overline{p}|\}^{\downarrow}$
B: 我 怀疑 他们 [有 这个 服务]$_{BF}$。
I $huaiyi_2$ they have this service
B: I suspect that they have this service. $[\![\varphi_1]\!]^o = \{|p|\}^{\downarrow}$
B': ?我 怀疑 他们 是不是 [有 这个 服务]$_{BF}$。
I $huaiyi_2$ they be-NEG-be have this service
B': I suspect that they have this service. $[\![\varphi_2]\!]^o = \{|p|, |\overline{p}|\}^{\downarrow}$
b. A: 这家 酒店 有 免费早餐。
This hotel have free breakfast
A: "This hotel provides free breakfast." $[\![\psi_2]\!]^o = \{|p|\}^{\downarrow}$
B: ?我 怀疑 他们 [有 这个 服务]$_{BF}$。
I $huaiyi_2$ they have this service
B: I suspect that they have this service. $[\![\varphi_3]\!]^o = \{|p|\}^{\downarrow}$
B': ?我 怀疑 他们 是不是 [有 这个 服务]$_{BF}$。
I $huaiyi_2$ they be-NEG-be have this service
B': I suspect that they have this service. $[\![\varphi_4]\!]^o = \{|p|, |\overline{p}|\}^{\downarrow}$

It can be observed that responses with o-values of their complements equal to or weaker (containing more information states) than that of the proposition added to the conversation table are perceived as less acceptable by several native speakers, as they found them redundant in the light of context. This motivates the construction of the second disjunct of presupposition (i) in the current entry. But same as in 4.2.1(IV), the observation is now only based on judgments from a very limited number of native speakers. Larger-scale interviews are necessary to confirm the observations and determine if the constructed presupposition (i) is reasonable.

(IV) **Embedding $A_{LT}Q$, $A_{LT}Q_VN$ and constituent *wh*-complements**
$huaiyi_2$ cannot embed $A_{LT}Q$, $A_{LT}Q_VN$ and constituent *wh*-complements. The following group of examples illustrates cases for $A_{LT}Q$ and $A_{LT}Q_VN$ complements:

(25) a. 我怀疑 他 是[广东人 -或者- 福建人]$_{BF}$。 ($P_{OL}Q$)
I $huaiyi_2$ he is Cantonese -or- Fujianese
"I suspect he is a Cantonese-or-Fujianese."
b. *我怀疑 他 是[广东人$^{\uparrow}$ 还是福建人$_{\downarrow}$]$_{BF}$。 ($A_{LT}Q$)
I $huaiyi_2$ he is Cantonese or Fujianese
Intended: I suspect whether he is a Cantonese or Fujianese.
c. *我怀疑 他 是[广东人 还是不是广东人]$_{BF}$。 ($A_{LT}Q_{vN}$)
I $huaiyi_2$ he is Cantonese or not a Cantonese
Intended: I suspect whether he is a Cantonese or not.

As in 3.2(III) and 4.2.1(V), this can be accounted for by contradictions arise during the application of the current entry.

As for $huaiyi_2$ + constituent *wh-* complements, sentences with this combination are also generally grammatical, but again, complements are with existential declarative interpretations:

(26) a. 我怀疑 [什么 气体泄漏了]$_{BF}$。
I $huaiyi_2$ what gas leak PERF
#"I suspect what gas has leaked."
"I suspect that some gas has leaked."

b. 门 是 开的，我怀疑 [谁 出去了]$_{BF}$。
The door be open, I $huaiyi_2$ who out PERF
* "The door is open, I suspect who has gone out."
"The door is open, I suspect someone has gone out."

c. 空气 很 潮湿，我怀疑 [哪里 下雨了]$_{BF}$。
The air very wet, I $huaiyi_2$ where rain PERF
"The air is wet, I suspect where has it rained."
"The air is wet, I suspect that it has rained somewhere."

d. 我怀疑 [有 哪个 机构 暗中 资助 他们]$_{BF}$。
I $huaiyi_2$ have which institution secretly fund them
#"I suspect which institution is funding them secretly."
"I suspect that some institution is funding them secretly."

Similarly, findings about quexistentials in Mandarin can be applied for the analysis here, and the interrogative readings of *wh*-complements are blocked due to the triviality caused by the type-mismatch.

(V) $P_{OL}Q$ **complements in negative contexts**
The negation of $huaiyi_2$ also seems to fuzzily accept B(e)-not-B(e) interrogative complements for some native speakers, although a supplementary statement following it can serve to make it clear. With this insight, the predicted contradiction arises from the presupposition (ii) and the assertive component under negation (which requires the doxastic state to be the subset of one alternative and doesn't intersect the other) may be undesired. This is revealed in the following example:

(27) 我不怀疑 小唐 是不是 [$_F$喜欢] 小强。(我觉得他们 压根 不熟)。
I don't $huaiyi_2$ Tang be-NEG-be like Qiang (I feel they totally NEG familiar)
"I don't suspect that Tang likes Qiang. (As I feel that they are totally not familiar with each other.)"

Similarly, this would require larger-scale interviews with native speakers to confirm the linguistic intuitions and determine if the status of presupposition (ii) under negation should be reconsidered.

5 Conclusions and future directions

This paper firstly provides an organized review on classical pieces of research on the semantics of dubitative predicates from both aspects of lexical semantic interpretations and complement selectional restrictions. It is noted that most research focus on one of the aspects, but with foundations they laid, comprehensive proposals for the lexical entry of dubitative predicates linking their lexical semantics and selectional restrictions is a desirable goal.

Therefore, the paper then thoroughly reviews Uegaki (2021), which is a recent research on English *doubt* in this comprehensive fashion. How this comprehensive analysis resolves a number of peculiar empirical problems left open by previous research is analyzed with a paradigm leaning more to Inquisitive Semantics, as this framework is one of the most important tools utilized in the research.

Lastly, the paper makes efforts to extend the analysis in Uegaki (2021) to the case of dubitative predicate 怀疑(huái yí) in Mandarin Chinese, construct alternative lexical entries in its fashion and make comparisons with the previous framework proposed in Huang (2020) during the process. Overall, new entries for both $huaiyi_1$ and $huaiyi_2$ are with a stronger power to account for selectional restrictions compared to the previous framework, especially regarding the compatibility with declarative/$P_{OL}Q$ complements and incompatibility with $A_{LT}Q$, $A_{LT}Q_VN$ and constituent *wh*-complements.

However, alternative entries also show limitations when dealing complications and empirical problems more specific to Mandarin Chinese. With insights from this process, some of future directions are proposed as follows:

- Carry out tests regarding nuances of acceptability between 怀疑(huái yí) + declarative complement and 怀疑(huái yí) + $P_{OL}Q$ complements in different contexts (with different conversation tables) with more native speakers for both interpretations of $huaiyi_1$ and $huaiyi_2$. Potentially, presuppositional components in alternative entries need to be adjusted based on the testing results;
- Search for a reasonable mechanism that can predict the interpretation in negative contexts and the preferential comparison at the same time for $huaiyi_2$ interpretation;
- Collect more data from native speakers to confirm the judgment regarding negation of 怀疑(huái yí) + $P_{OL}Q$ complements and determine if presuppositional components in alternative entries need to be adjusted;
- If the issues stated above can be successfully resolved, come up with a unified semantics of 怀疑(huái yí) that can satisfactorily explain its selectional restrictions in a meaning-driven way;
- Combine experimental methods and computational techniques to verify the interpretation strategy and the usage of 怀疑(huái yí) to collect more evidence from "language in real use".

References

Anand, P., & Hacquard, V. (2013). Epistemics and Attitudes. *Semantics and pragmatics*, *6*, 1–59. https://doi.org/10.3765/sp.6.8

Bar-Lev, M. E. (2020). An implicature account of homogeneity and non-maximality. *Linguistics and Philosophy*, *44*. https://doi.org/10.1007/s10988-020-09308-5

Bar-Lev, M. E., & Fox, D. (2020). Free choice, simplification, and innocent inclusion. *Natural Language Semantics*, *28*. https://doi.org/10.1007/s11050-020-09162-y

Biezma, M. (2009). Alternative vs Polar Questions: the Cornering Effect. *Proceedings of SALT 19*, 37–54. https://journals.linguisticsociety.org/proceedings/index.php/SALT/article/download/2519/2267

Biezma, M., & Rawlins, K. (2012). Responding to Alternative and Polar Questions. *Linguistics and Philosophy*, *35*(5), 361–406. http://www.jstor.org/stable/41811566

Bolinger, D. (1968). Post-posed Main Phrases: An English Rule for the Romance Subjunctive. *Canadian Journal of Linguistics*, *14*, 3–30. https://doi.org/10.1017/S0008413100019629

Center for Chinese Linguistics PKU. (2023). CCL Online Corpus. http://ccl.pku.edu.cn:8080/ccl_corpus/

Chierchia, G., Fox, D., & Spector, B. (2013). Scalar implicature as a grammatical phenomenon. In K. v. H. Claudia Maienborn & P. Portner (Eds.), *Semantics: An international handbook of natural language meaning* (pp. 2297–2332, Vol. 3). De Gruyter Mouton.

Ciardelli, I., Groenendijk, J., & Roelofsen, F. (2019a). Disjunction, clause typing, and inotation. In *Inquisitive semantics* (pp. 93–113). Oxford University Press. https://fdslive.oup.com/www.oup.com/academic/pdf/openaccess/9780198814788.pdf

Ciardelli, I., Groenendijk, J., & Roelofsen, F. (2019b). *Inquisitive semantics*. Oxford University Press. https://fdslive.oup.com/www.oup.com/academic/pdf/openaccess/9780198814788.pdf

Ciardelli, I., Groenendijk, J., & Roelofsen, F. (2019c). Questions. In *Inquisitive semantics* (pp. 77–92). Oxford University Press. https://fdslive.oup.com/www.oup.com/academic/pdf/openaccess/9780198814788.pdf

Dixon, R. M. (2005). *A semantic approach to english grammar (2nd edition)*. Oxford University Press.

Fischer, M. (2005). Ein zweifelsfall: *Zweifeln* im deutschen. *Linguistische Berichte*, *202*, 127–169. https://doi.org/10.21248/zaspil.30.2003.179

Groenendijk, J., & Stokhof, M. (1984). *Studies on the semantics of questions and pragmatics of answers* [Doctoral dissertation, University of Amsterdam]. https://dare.uva.nl/search?identifier=6abca507-02b1-4df1-b643-78a8c249f43c

Hagstrom, P. (2006). A-not-A Questions. In M. Everaert & H. van Riemsdijk (Eds.), *The Blackwell Companion to Syntax, Volume I* (pp. 173–213). Blackwell Pub.

Hamblin, C. L. (1973). Questions in Montague English. *Foundations of Language*, *10*, 41–53. http://www.jstor.org/stable/25000703

Han, L. (2001). On the Sense, Object and Sentence Meaning of *huaiyi*. (Chinese). *Journal of Xuzhou Normal University (Philosophy and Social Sciences)*, *Vol.27*(1), 35–38. https://doi.org/10.16095/j.cnki.cn32-1833/c.2001.01.007

Hengeveld, K., Iatridou, S., & Roelofsen, F. (2022). Quexistentials and Focus. *Linguistic Inquiry*, 1–54. https://doi.org/10.1162/ling_a_00441

Hinterwimmer, S. (2012). Information Structure and Truth-Conditional Semantics. In C. M. Klaus von Heusinger & P. Portner (Eds.), *Semantics: An International Handbook of Natural Language Meaning, Volume 2*. De Gruyter Mouton. https://doi.org/10.1515/9783110255072.1875

Huang, Y. (2020). *What does huaiyi mean?: An investigation into the semantics of the dubitative verb in mandarin chinese* [Master's thesis]. School of Philosophy, Psychology and Language Sciences, University of Edinburgh. https://www.dropbox.com/s/4ine28p226be6dr/Yunchong%20Huang%20-%20What%20does%20huaiyi%20mean.pdf?dl=0

Karttunen, L. (1977a). Syntax and Semantics of Questions. *Linguistics and Philosophy*, *1*(1), 3–44. https://doi.org/10.1007/BF00351935

Karttunen, L. (1977b). To doubt whether. In *The CLS book of squibs*. Chicago Lingsuitics Society.

Li, X. (1987). The Meaning of *huaiyi* and the Types of its Obejcts. (Chinese). *Chinese Language*, *2*, 114–117.

Liu, B., & Yuan, Y. (2021). On the distribution and meaning construcal of huaiyi under the theory of information structure and conceptual polarity (chinese). *Journal of Foreign Languages*, *44*(5), 13–21. http://jfl.shisu.edu.cn/CN/abstract/abstract577.shtml#1

Liu, M., & Yang, Y. (2021). Modal wh-indefinites in mandarin. *Sinn und Bedeutung 25*. https://doi.org/10.18148/sub/2021.v25i0.955

Lu, C. (2016). A New Study on the Extension Mechanism and Construal Strategy of the Sense of the Chinese Verb *huaiyi*: Also on the Syntactic Influence of the Game between "the Least Effort Principle of Expressions" and "the Least Effort Principle of Understanding". (Chinese). *Language Teaching and Research*, *179*(3), 93–101.

Pruitt, K., & Roelofsen, F. (2011). Disjunctive Questions: Prosody, Syntax, and Semantics. http://projects.illc.uva.nl/inquisitivesemantics

Roelofsen, F., & Farkas, D. F. (2015). Polarity particle responses as a window onto the interpretation of questions and assertions. *Language*, *91*(2), 359–414.

Rooth, M. (1992). A Theory of Focus Interpretation. *Natural Language Semantics*, *1*(1), 75–116. http://www.jstor.com/stable/23748778

Rooth, M. (2016). Alternative Semantics. In C. Féry & S. Ishihara (Eds.), *The Oxford Handbook of Information Structure*. Oxford University Press. https://doi.org/10.1093/oxfordhb/9780199642670.013.19

Rooth, M. (1985). *Association with Focus* [Doctoral dissertation, GLSA, University of Massachusetts, Amherst.].

Schwarzschild, R. (1999). Givenness, AVOIDF and Other Constraints on the Placement of Accent. *Natural Language Semantics*, *7*, 141–177. http://www.jstor.org/stable/23748636

Uegaki, W. (2021). The doubt-whether puzzle. In Ł. Jdrzejowski & C. Umbach (Eds.), *Non-interrogative subordinate wh-clauses* (pp. 461–491). Oxford University Press. https://doi.org/10.31234/osf.io/7ryzu

Uegaki, W., & Sudo, Y. (2019). The **hope-wh* puzzle. *Natural Language Semantics*, *27*. https://doi.org/10.1007/s11050-019-09156-5

Yuan, Y. (2014). On the Sense Extension Mechanism and Semantic Construal Strategy of the Chinese Verb *huaiyi*. (Chinese). *Studies in Language and Linguistics*, *34*(3), 1–12.

Zhang, J., & Wu, J. (2013). A Multi-dimensional Interpretation of Chinese Antagonymic Verb *huaiyi*. (Chinese). *Journal of Zhejiang International Studies University*, *6*, 1–8.